EATR-Stereo: Embodiment-Aware Routing of Paired Stereo Evidence for Humanoid Vision-Language-Action Control

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***Abstract*—Long-horizon humanoid vision–language–action (VLA) control with head-mounted stereo cameras requires visual interfaces that can exploit complementary views while maintaining compatibility with pretrained representations. Existing interfaces often discard complementary stereo evidence or fuse additional observations without preserving the native primary-view pathway and adapting auxiliary information to robot embodiment. We present EATR-Stereo, an embodiment-aware token-routing framework that retains primary-view tokens and constructs primary-aligned Cross-View Auxiliary Tokens (CVATs) by querying the synchronized auxiliary-view token sequence. A body-segmented proprioceptive encoder further conditions token-wise auxiliary usage on robot configuration history, enabling selective incorporation of stereo evidence during action generation. The routed auxiliary stream augments the language and primary-visual context of a pretrained VLA while keeping its vision–language model frozen. On a 33-DoF physical humanoid with a 37-D proprioceptive state, we evaluate nine configurations in over-100-s search–approach–grasp–place–return tasks. EATR-Stereo achieves 60.0% full-task success, 100.0% grasp success, and 80.0% stage success. Under severe asymmetric occlusion, it improves recovery to 80% compared with 30% for CVAT alone. Ablation studies further show the importance of preserving primary tokens and combining cross-view auxiliary features with structured proprioceptive routing. These results demonstrate that selectively routed paired stereo evidence improves spatial grounding for reliable long-horizon humanoid VLA control.**

***Index Terms*—Humanoid robots, vision–language–action models, stereo vision, multimodal fusion, long-horizon manipulation**

I. INTRODUCTION

Vision–language–action (VLA) models transfer large-scale vision–language representations to language-conditioned control [1], [2], and recent systems extend them to continuous actions and whole-body humanoid behavior [3], [4]. Bipedal deployment, however, couples perception to whole-body motion: gait and torso disturbances reach head-mounted cameras, head and waist motion alter viewpoint, and moving arms and hands create configuration-dependent self-occlusion [5], [6]. In long-horizon tasks, missed evidence at one stage can propagate through the remainder.

Synchronized stereo provides complementary evidence for this regime. Its cameras observe the same instant from a fixed or calibratable baseline, supporting correspondence and disparity while providing complementary visibility, so content occluded in one view may remain visible in the other [7], [8]. Unlike arbitrary multi-view imagery, stereo pairs share timing, baseline, and scene content; the useful signal is therefore the paired relation, not merely a second image.

Current VLA interfaces exploit only part of this signal. A monocular input lacks binocular disparity and an alternate view of occluded regions [7]; independent token concatenation retains both images but does not enforce cross-view-consistent spatial reasoning [8]. StereoPolicy explicitly relates the views, but its VLA variant replaces the original image tokens with a fused stereo representation, discarding the pretrained monocular token structure [7]. SpatialVLA instead predicts depth from RGB with ZoeDepth and back-projects pixels into camera-frame 3D, while its reported physical evaluations use arm manipulators with static or tripod-mounted cameras [9]. They therefore do not cover head-mounted bipedal perception, where monocular depth may be temporally inconsistent and gait disturbs the cameras [6], [10], [11]. Existing visual–proprioceptive fusion likewise does not condition paired-view selection on body-semantic state despite configuration-dependent viewpoint and occlusion [7], [8], [12], [13].

We therefore propose **EATR-Stereo**, an Embodiment-Aware Token Routing framework for pretrained humanoid VLAs. It preserves primary-view tokens, queries the synchronized auxiliary view to construct primary-aligned Cross-View Auxiliary Tokens (CVATs), and uses body-segmented proprioceptive history to predict token-wise routing weights. The routed auxiliary stream augments the original language and primary-visual context before the shared continuous-action expert, while the vision–language model remains frozen. Fig. 1 summarizes the paired sensing setting and interface requirements.

Our contributions are:

- A paired-stereo VLA interface that constructs primary-aligned CVATs while preserving the pretrained primary representation.
- Embodiment-aware, token-wise routing conditioned on body-segmented proprioceptive history and cross-view relations.
- A nine-configuration study on a 33-DoF humanoid in over-100-s tasks. EATR-Stereo reaches 60.0

II. RELATED WORK

A. Vision–Language–Action Models

Vision–language–action (VLA) models leverage pretrained vision–language representations to improve semantic generalization in robot control. Early representative approaches

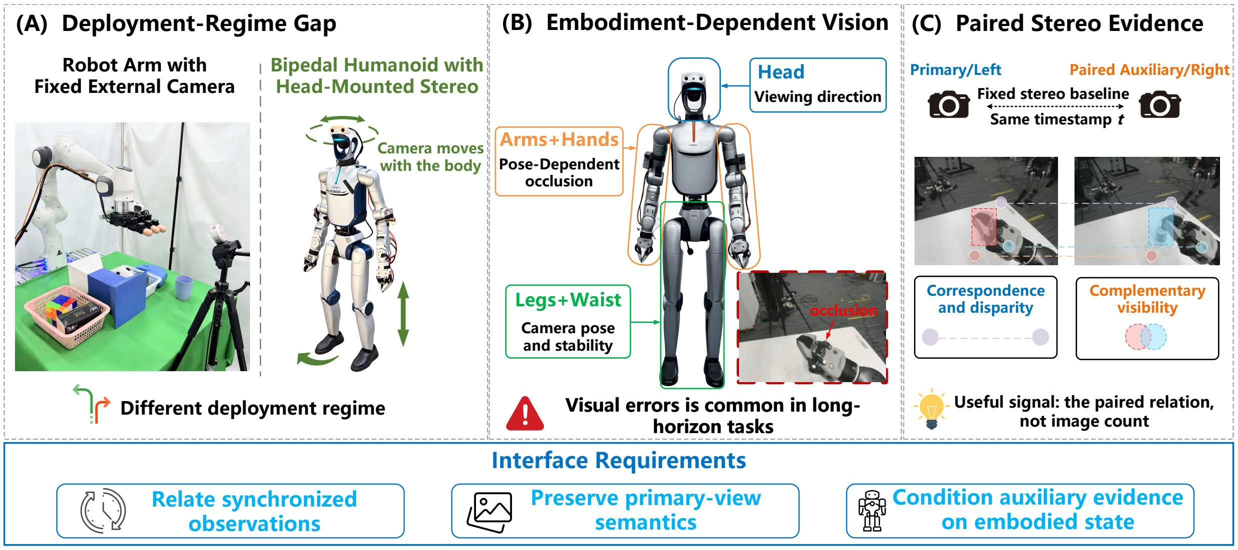

Fig. 1. Synchronized stereo evidence and visual-interface comparison. A synchronized pair couples correspondence and disparity with complementary visibility. Independent token concatenation retains both views but does not explicitly relate them, while StereoPolicy [7] produces a fused stereo representation. EATR-Stereo preserves the primary semantic tokens, constructs a separate primary-aligned CVAT stream, and routes this auxiliary evidence with body-segmented proprioception before the shared flow-matching action head.

either cast actions into the token space or learn generalist policies across heterogeneous robot data, as exemplified by RT-2, OpenVLA, and Octo [1], [2], [14]. More recent VLAs increasingly decouple semantic perception and reasoning from continuous action generation, conditioning diffusion- or flow-based action experts on VLM features [3], [15]. This paradigm has further been extended through heterogeneous co-training and high-level semantic prediction in $\pi_{0.5}$ [16], as well as toward multi-embodiment and whole-body humanoid control in GR00T N1 and LingBot-VLA 2.0 [4], [17]. Despite these advances in semantic transfer, action generation, and embodiment generalization, existing VLA architectures largely treat visual observations as a conventional semantic input stream and do not explicitly exploit the correspondence and complementary visibility available from synchronized stereo views.

B. Stereo and Multi-View Robot Learning

Robot policies incorporate spatial information through explicit 3D representations, depth-enhanced visual features, and multi-view observations. Explicit geometric approaches operate on point clouds or rendered 3D views [18]–[20], while depth-based methods augment 2D representations with monocular geometric priors [9], [21]. Multi-view policies instead seek complementary spatial evidence across observations, including geometry-grounded task inputs and synthesized auxiliary views [22], [23]. Most closely related to our setting, StereoPolicy processes synchronized stereo images with shared 2D encoders and cross-view attention, enabling implicit geometric reasoning without explicit disparity or point-cloud reconstruction [7]. EATR-Stereo differs in how stereo evidence is integrated into a pretrained VLA: it retains the native primary-view representation and appends a primary-aligned auxiliary stream, allowing complementary cross-view evidence to be selectively incorporated without replacing the original semantic pathway.

C. Reliability-Aware and Proprioception-Conditioned Fusion

Robot policies must integrate visual evidence with proprioceptive state while remaining robust to changes in observation reliability. Existing approaches either align heterogeneous vision and state inputs within a shared representation space [12], adaptively rebalance their relative contributions [13], or improve robustness to corrupted observations through filtering and perturbation-based training [24], [25]. These methods primarily address cross-modal alignment or observation robustness at the representation level. EATR-Stereo instead uses structured proprioceptive history as a routing condition for cross-view evidence: body-segmented state features are combined with primary–auxiliary visual relations to modulate auxiliary stereo tokens, enabling state-dependent selection of complementary visual information while preserving the primary semantic stream.

In summary, EATR-Stereo focuses on synchronized stereo rather than generic multi-view input, preserves the pretrained primary representation while appending aligned auxiliary evidence, and routes this auxiliary pathway with structured proprioceptive context rather than relying on static fusion.

III. METHODS

A. Problem Formulation

We formulate long-horizon humanoid manipulation as the prediction of continuous action chunks conditioned on synchronized stereo observations, language instructions, and proprioceptive history. At time t, the policy observation is

$$o_t = \left(I_t^L, I_t^R, \ell_t, S_t\right), \qquad S_t = (s_{t-K+1}, \ldots, s_t), \tag{1}$$

where (I_t^L, I_t^R) is a synchronized stereo pair, ℓ_t is the current language instruction, and s_t denotes the robot proprioceptive state at a single control step. Thus, S_t contains the latest K proprioceptive observations.

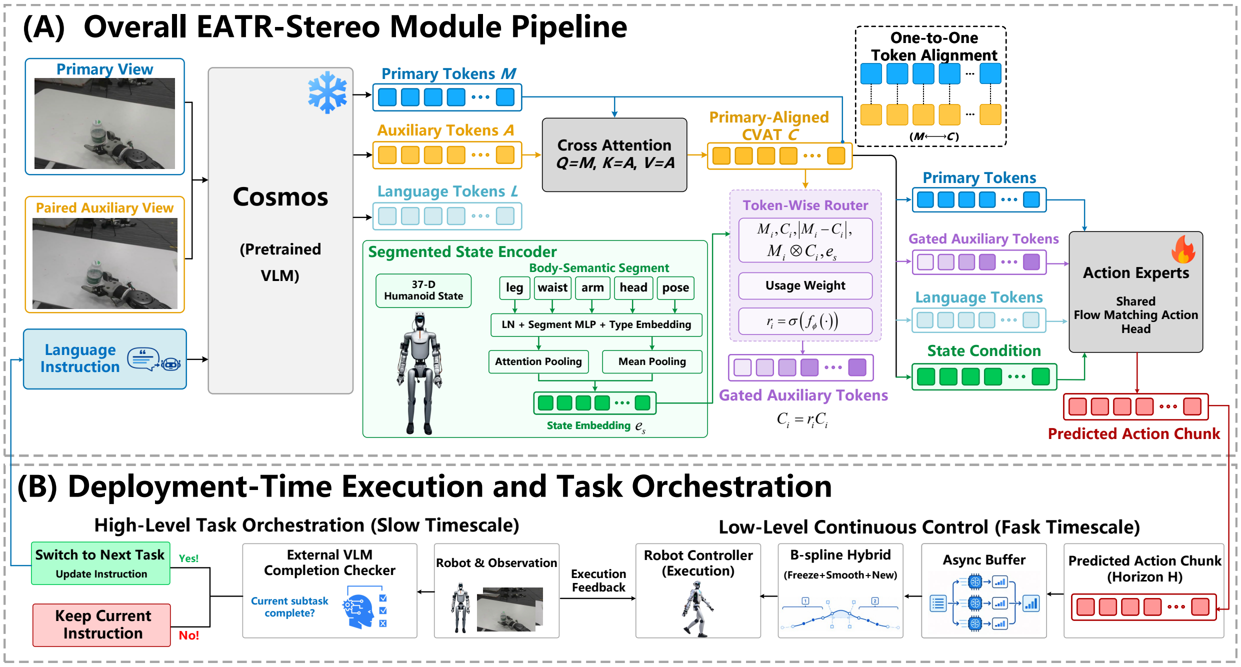

Fig. 2. EATR-Stereo architecture and deployment flow. In (A), the frozen Cosmos VLM jointly encodes the synchronized primary–auxiliary image pair and the language instruction. Cross-view attention constructs primary-query-aligned CVATs, while segmented proprioception conditions their token-wise routing before the shared flow-matching head predicts an action chunk. In (B), asynchronous buffering and local B-spline stitching support continuous control. An external vision–language model (VLM) supplies a subtask-completion signal used by a finite-state machine (FSM) to retain or advance the subtask and update the instruction.

The policy predicts an H-step continuous-action chunk

$$\widehat{\mathcal{Y}}_t = (\hat{\mathbf{a}}_t, \ldots, \hat{\mathbf{a}}_{t+H-1}) = \pi_\theta(I_t^L, I_t^R, \ell_t, S_t), \tag{2}$$

where $\hat{\mathbf{a}}_t$ is the predicted continuous robot control command at one execution step. We use $\mathcal{Y}_t = (\mathbf{a}_t, \ldots, \mathbf{a}_{t+H-1})$ to denote the corresponding target action chunk.

The two images are captured simultaneously by cameras with a fixed or calibratable baseline, providing correspondence and parallax cues as well as complementary visibility when scene content is occluded in one view. We designate one image as the primary view I_t^p and the other as the paired auxiliary view I_t^a, and keep this assignment fixed during training and deployment. EATR-Stereo selectively incorporates paired auxiliary evidence while retaining an independent primary-token pathway, as illustrated in Fig. 2(A).

B. Cross-View Auxiliary Token Construction

As illustrated in Fig. 2(a), the Cross-View Auxiliary Token (CVAT) module constructs a primary-indexed auxiliary visual stream while preserving the native primary-view tokens. The synchronized stereo pair and language instruction are jointly processed by the frozen Cosmos VLM [4]. The resulting hidden states are separated according to token origin into the language sequence L_t and the primary and auxiliary visual sequences,

$$M_t = E_{\mathrm{C}}(I_t^p), \qquad A_t = E_{\mathrm{C}}(I_t^a), \tag{3}$$

where $M_t, A_t \in \mathbb{R}^{N\times d}$ share the same VLM hidden space. Learned view-role embeddings are added to both sequences, after which the notation M_t and A_t is retained. The primary sequence M_t is retained unchanged along the native visual pathway, while the auxiliary sequence A_t is used to construct the CVAT stream.

To associate auxiliary evidence with individual primary tokens, the CVAT module applies multi-head cross-attention, following the cross-view fusion principle used in StereoPolicy [7], with the primary sequence as queries and the auxiliary sequence as keys and values. For attention head h,

$$\begin{aligned} Q_h &= \mathrm{LN}(M_t)W_h^Q, \\ K_h &= \mathrm{LN}(A_t)W_h^K, \\ V_h &= \mathrm{LN}(A_t)W_h^V, \end{aligned} \tag{4}$$

and the corresponding head output is

$$O_h = \mathrm{softmax}\left(\frac{Q_h K_h^\top}{\sqrt{d_h}}\right) V_h, \tag{5}$$

where d_h is the head dimension. Each primary query therefore adaptively aggregates evidence from the full synchronized auxiliary-token sequence.

We concatenate and project the outputs of all N_h heads and then apply a token-wise feed-forward residual block:

$$C_t^{(0)} = \mathrm{Concat}_{h=1}^{N_h}(O_h)W^O, \quad C_t = C_t^{(0)} + \mathrm{MLP}\left(\mathrm{LN}(C_t^{(0)})\right) \tag{6}$$

yielding $C_t \in \mathbb{R}^{N\times d}$. Since cross-attention produces one output for each primary query, C_t inherits the token indexing of M_t. Specifically, each $C_{t,i}$ is computed by attending from $M_{t,i}$ to the full auxiliary sequence A_t, establishing primary-indexed alignment between M_t and C_t. The resulting CVAT stream C_t is passed to the token router, while the primary stream M_t is retained unchanged.

C. Embodiment-Aware Routing via Segmented Proprioception

As illustrated in Fig. 2(A), the embodiment-aware router conditions token-wise CVAT usage on a structured representation of recent proprioceptive history. We partition the proprioceptive state into $G = 5$ semantic groups: legs, arms, head, waist, and robot pose, where the arms group includes both arm and hand states. The dimensionality of each group may vary across platforms.

For each segment g, its K-frame history is flattened, normalized by LayerNorm, and mapped to a common feature dimension by an independent lightweight MLP, followed by a learned segment-type embedding:

$$z_{t,g} = f_g\left(\mathrm{LN}_g\left(\mathrm{Flatten}(s_{t-K+1:t,g})\right)\right) + e_g, \quad g = 1, \ldots, G, \tag{7}$$

where $z_{t,g} \in \mathbb{R}^{d_s}$ denotes the resulting segment representation.

The segment representations are aggregated through attention-weighted and mean pooling:

$$\begin{aligned} \alpha_{t,g} &= \frac{\exp(f_{\mathrm{attn}}(z_{t,g}))}{\sum_{j=1}^{G} \exp(f_{\mathrm{attn}}(z_{t,j}))}, \\ h_t^s &= f_{\mathrm{fuse}}\left(\left[\sum_{g=1}^{G} \alpha_{t,g} z_{t,g}; \frac{1}{G}\sum_{g=1}^{G} z_{t,g}\right]\right). \end{aligned} \tag{8}$$

For each token index i, the router jointly uses the primary token $M_{t,i}$, the corresponding CVAT token $C_{t,i}$, their element-wise feature relationships, and the shared proprioceptive condition h_t^s. Specifically,

$$\begin{aligned} \bar{M}_{t,i} &= \mathrm{LN}(M_{t,i}), \qquad \bar{C}_{t,i} = \mathrm{LN}(C_{t,i}), \\ q_{t,i} &= \left[\bar{M}_{t,i}; \bar{C}_{t,i}; |\bar{M}_{t,i} - \bar{C}_{t,i}|; \bar{M}_{t,i} \odot \bar{C}_{t,i}; h_t^s\right], \end{aligned} \tag{9}$$

where $|\cdot|$ and $\odot$ denote element-wise absolute difference and product, respectively.

A lightweight routing MLP predicts one scalar gate for each CVAT token:

$$r_{t,i} = \sigma\left(f_r(q_{t,i})\right), \qquad \widetilde{C}_{t,i} = r_{t,i} C_{t,i}, \tag{10}$$

where $r_{t,i} \in (0,1)$ and $\widetilde{C}_t \in \mathbb{R}^{N \times d}$. The router scales only the CVAT stream: M_t contributes to the routing descriptor but remains unmodified. The routed auxiliary tokens $\widetilde{C}_t$ and the primary tokens subsequently meet only in the context supplied to the action experts.

D. VLA Integration, Training, and Deployment

As shown in Fig. 2(A), the routed auxiliary tokens are appended to the language-token sequence and the preserved primary-token sequence:

$$E_t^{\mathrm{VLA}} = \left[L_t; M_t; \widetilde{C}_t\right], \tag{11}$$

The resulting context is supplied to the GR00T N1.7 action experts together with the original robot-state condition S_t; the segmented state representation introduced in Sec. III-C is used only for auxiliary-token routing and does not replace this state pathway. Following the standard GR00T N1.7 formulation [4], the action experts are trained with the flow-matching objective

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,\tau}\left[\left\| v_\theta\left(\mathcal{Y}_t^{\tau}, \tau \mid E_t^{\mathrm{VLA}}, S_t\right) - u_\tau \right\|_2^2\right], \tag{12}$$

where $\mathcal{Y}_t^{\tau}$ is the interpolated state of target action chunk $\mathcal{Y}_t$ at flow time τ, u_τ is the target velocity field, and v_θ is the predicted field. The Cosmos VLM remains frozen, while the CVAT module, segmented state encoder, token router, and GR00T action experts are jointly optimized.

During deployment, action chunks are predicted asynchronously and placed in an execution buffer. At each transition, the hybrid trajectory retains the committed prefix, replaces a local window with a cubic B-spline, and appends the untouched suffix of the new chunk. The transition spline is

$$B(\eta) = \sum_{i=0}^{n} N_{i,3}(\eta)\mathbf{c}_i, \tag{13}$$

where η is local time, $N_{i,3}$ is the cubic B-spline basis, and $\{\mathbf{c}_i\}_{i=0}^{n}$ are the $n+1$ action-space coefficient vectors. The spline matches the action and velocity at the stitching boundary and inherits its terminal value and finite-difference velocity from the new transition window. A short cubic-Hermite adjustment precedes uniform waypoint sampling, including the window endpoint, and independent spline fitting along each action dimension.

At the task level, an external VLM evaluates subtask completion from visual observations and outputs a binary completion signal. The FSM retains the current instruction until completion and then advances to the next subtask, whose instruction is encoded into L_t for subsequent policy inference. The external VLM is used only for subtask-completion evaluation.

IV. EXPERIMENTS

A. Experimental Questions and Setup

We evaluate EATR-Stereo along four dimensions: (i) long-horizon physical-humanoid performance, (ii) contributions of its representation and routing components, (iii) generalization to position shifts and language-conditioned target selection, together with robustness to severe asymmetric occlusion, and (iv) whether the relative method ranking transfers to larger-scale arm-based simulation.

Training and comparisons: For the physical-humanoid experiments, all nine configurations use the same GR00T N1.7 backbone [4], 1000 segmented-task demonstrations, and identical optimization settings. The VLM is frozen; each policy uses a six-frame state history and is trained for 60,000 steps with a global batch size of 512 on 16 NVIDIA H20 GPUs. The baselines include GR00T1.7-Mono, GR00T-Wide, the default two-image token-concatenation GR00T [4], and StereoPolicy [7] adapted to the same backbone. We additionally evaluate Main-XAttn, CVAT, CVAT-Flat, Stereo-Route, and EATR-Stereo. Main-XAttn updates the primary stream through cross-attention, whereas CVAT preserves the native primary tokens and appends a primary-aligned auxiliary stream. CVAT-Flat and EATR-Stereo differ only in flat versus body-segmented proprioceptive encoding, while Stereo-Route replaces CVAT with the stereo-fusion scheme of StereoPolicy.

Simulation protocol: We conduct 3600 Franka-arm rollouts across 18 RoboCasa365 tasks [26], with 20 trials per task and configuration. QwenPI0.5 [16] is included only in simulation as an additional external VLA baseline. All policies are trained from RoboCasa365 demonstrations under the same protocol.

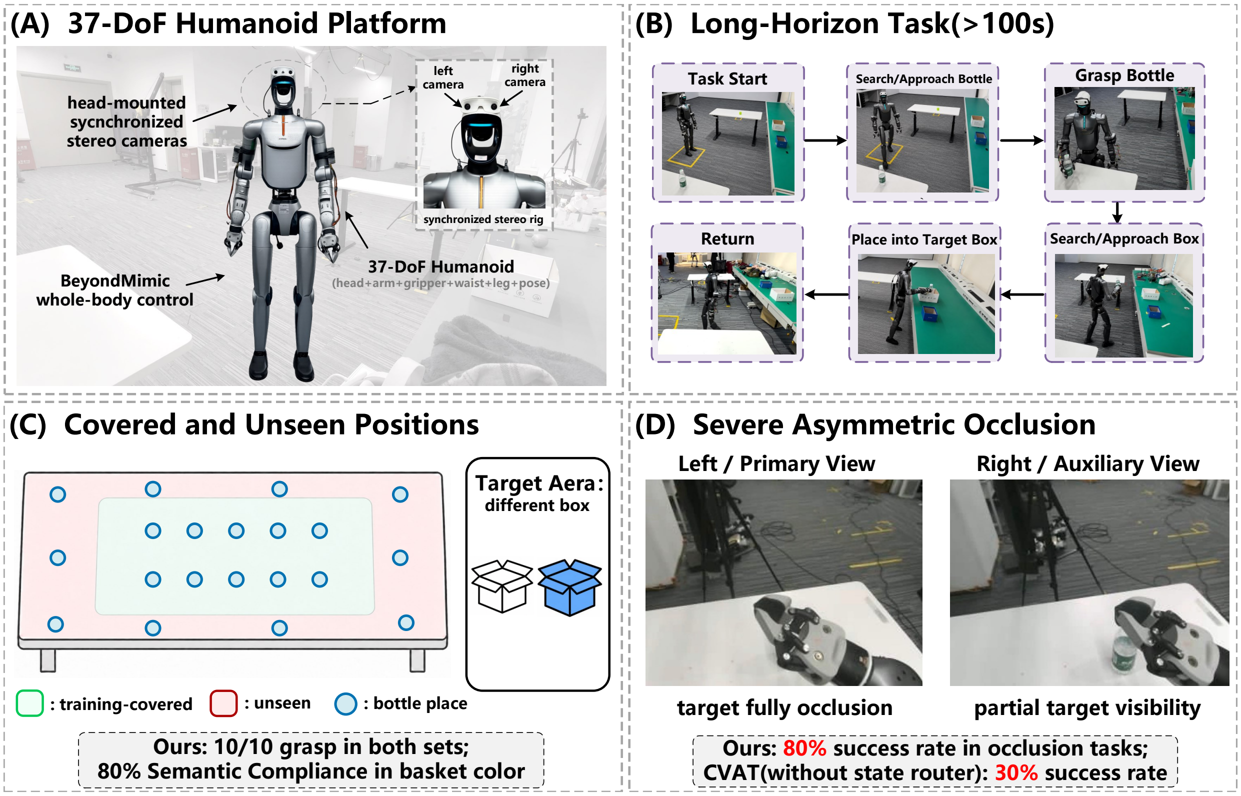

Fig. 3. Physical-humanoid evaluation. (A) The 37-DoF platform and synchronized head-mounted stereo rig. (B) The over-100-s search–approach–grasp–place–return task. (C) Relative layout of demonstration-covered and unseen bottle positions; absolute coordinates are not used in the comparison. (D) Severe asymmetric occlusion, where the bottle is fully hidden in the primary view but remains partially visible in the auxiliary view.

Wrist-camera observations are excluded, leaving only synchronized left and right agent views to isolate paired-view modeling. Aggregate task success is the primary simulation metric.

Physical-humanoid protocol: Experiments are conducted on a 33-DoF HONOR Omega 1.0 humanoid platform, controlled by BeyondMimic [27]. The 37-D proprioceptive state comprises the 33 joint dimensions and a 4-D base-orientation quaternion. The head-mounted stereo rig provides synchronized 240×320 images, with left and right views used as primary and auxiliary inputs without camera parameters. Policies run at 10 Hz on an NVIDIA RTX 4090 and predict 30-step action chunks. All methods share the same local cubic B-spline chunk transition ($k = 3$), using eight sampled waypoints and eight blending steps.

Each complete task lasts more than 100 s and consists of bottle search, approach, grasp, language-conditioned placement into the instructed colored basket, and return to the initial position [Fig. 3(B)]. We conduct 180 physical trials, with 20 per configuration, evenly split between demonstration-covered (ID) and unseen (OOD) bottle positions [Fig. 3(C)]. Objects are reset before each trial, and all methods share the same external vision–language monitor for binary subtask-completion detection and finite-state instruction switcher.

For severe asymmetric occlusion [Fig. 3(D)], CVAT, CVAT-Flat, and EATR-Stereo are each evaluated over 10 trials with a 90-s timeout. The robot hand fully occludes the bottle in the primary view while leaving it partially visible in the auxiliary view. Recovery requires a successful lift within the timeout, with recovery time measured from placement at the occluded position to the successful lift.

TABLE I
Top-seven RoboCasa365 results over all 18 tasks.

Method	Success (%) ↑	Success/Trials
EATR-Stereo	**43.33**	**156/360**
CVAT	39.44	142/360
StereoPolicy	38.06	137/360
CVAT-Flat	37.50	135/360
QwenPI0.5	36.11	130/360
GR00T1.7	35.55	128/360
GR00T1.7-Mono	32.77	118/360

Metrics: Full-task success requires grasping the bottle, placing it in the instructed basket, and returning to the initial position; grasp success requires a successful lift. Stage success is the mean of grasp and full-task success, with ID and OOD rates computed separately over their respective 10-trial subsets. Semantic compliance measures correct selection of the language-specified basket color. Full-task success is the primary physical-world metric.

B. RoboCasa Arm-Based Simulation Validation

Before the primary physical-humanoid evaluation, we conduct a large-scale arm-based simulation study on RoboCasa365 to provide preliminary validation over a broader task set. Table I reports aggregate success across all 18 tasks.

TABLE II
MAIN PHYSICAL-HUMANOID COMPARISON OVER 20 TRIALS PER METHOD. ID AND OOD DENOTE DEMONSTRATION-COVERED AND UNSEEN BOTTLE POSITIONS, RESPECTIVELY.

Method	Full-task Success ↑	Grasp Success ↑	Stage Success Rate ↑	ID Stage Success ↑	OOD Stage Success ↑
GR00T1.7-MONO	35.0%	55.0%	45.0%	55.0%	35.0%
GR00T-WIDE	20.0%	45.0%	32.5%	40.0%	25.0%
GR00T	35.0%	80.0%	57.5%	75.0%	40.0%
StereoPolicy	45.0%	85.0%	65.0%	80.0%	50.0%
EATR-Stereo	**60.0%**	**100.0%**	**80.0%**	**90.0%**	**70.0%**

EATR-Stereo achieves the highest aggregate success rate of 43.33% (156/360), outperforming CVAT, StereoPolicy, CVAT-Flat, QwenPI0.5, GR00T1.7, and GR00T1.7-MONO by 3.89, 5.27, 5.83, 7.22, 7.78, and 10.56 percentage points (pp), respectively. QwenPI0.5 is included only in the simulation study as an additional VLA baseline, instantiating the $\pi_0.5$ framework [16] with a Qwen-2B [28] VLM backbone. The relatively low absolute success rates should be interpreted in light of the restricted observation setting: unlike the original RoboCasa setup [26], wrist-camera observations are excluded, while the paired agent views exhibit substantial occlusion in several tasks. Despite this more challenging visual setting, EATR-Stereo consistently leads the compared methods, providing large-scale simulation support for the proposed cross-view representation and state-conditioned routing before their evaluation on the physical humanoid platform.

C. Real-Humanoid Main Comparison

The primary evaluation measures long-horizon physical-humanoid manipulation under whole-body motion, dynamic self-occlusion, and unseen target configurations. Table II compares EATR-Stereo with three GR00T-based baselines [4] and StereoPolicy [7] in terms of full-task, grasp, and stage success, with stage success further separated into demonstration-covered (ID) and unseen (OOD) positions.

EATR-Stereo leads every metric in Table II. It completes 12 of 20 trials end-to-end (60.0%), outperforming StereoPolicy by 15.0 pp, GR00T1.7-MONO and GR00T by 25.0 pp, and GR00T-WIDE by 40.0 pp. It also grasps the bottle in all 20 trials, a 15.0–55.0 pp improvement over the baselines. The lower success of GR00T1.7-MONO mainly arises during approach, where the robot more frequently deviates from the bottle-directed walking trajectory. In contrast, EATR-Stereo maintains more accurate bottle-directed locomotion, suggesting that paired-view integration provides stronger spatial grounding for humanoid navigation and manipulation. GR00T-WIDE performs worst overall, which likely reflects the frozen VLM's limited pretraining exposure to wide-aspect-ratio images.

Its overall stage success is 80.0%, versus 65.0% for StereoPolicy and 32.5–57.5% for the GR00T-based baselines. EATR-Stereo achieves 90.0% and 70.0% stage success on ID and OOD positions, respectively; its 20.0 pp OOD advantage over StereoPolicy indicates stronger generalization to unseen bottle positions.

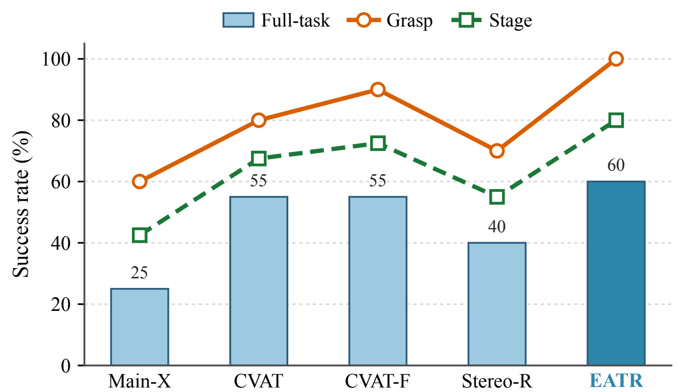

Fig. 4. Physical-humanoid ablations over 20 trials per method. CVAT-F denotes CVAT-Flat and Stereo-R denotes Stereo-Route.

D. Real-Humanoid Ablations and Targeted Analyses

Component ablations: We ablate three design choices: retaining the native primary-view stream, selectively routing auxiliary-view evidence, and encoding proprioceptive history by body-semantic segments. Fig. 4 reports full-task success as bars and grasp and stage-success rates as curves.

Replacing direct modification of the primary tokens (Main-XAttn) with the CVAT dual-stream representation raises full-task success from 25.0% to 55.0%, a 30.0 pp gain (Fig. 4). This result supports retaining the pretrained primary representation while introducing auxiliary evidence through a dedicated stream.

CVAT and CVAT-Flat both achieve 55.0% full-task success, but flat state conditioning improves grasp and stage success from 80.0% to 90.0% and from 67.5% to 72.5%, respectively. Body-segmented conditioning further increases these metrics to 60.0%, 100.0%, and 80.0% for full-task, grasp, and stage success. Stereo-Route achieves 40.0% full-task and 55.0% stage success, while EATR-Stereo further improves these metrics through the joint use of preserved primary tokens, the CVAT auxiliary stream, and body-segmented proprioceptive routing. Overall, the ablations show complementary gains from representation preservation, auxiliary-view integration, and structured state conditioning.

Position and semantic behavior:

Table III evaluates position-wise grasp generalization and language compliance. EATR-Stereo is the only method with 100% grasp success on both ID and OOD positions. Its OOD grasp rate exceeds CVAT-Flat by 20.0 pp, StereoPolicy by 30.0 pp, and CVAT and GR00T by 40.0 pp. Its zero ID–OOD

TABLE III
POSITION GENERALIZATION AND LANGUAGE COMPLIANCE ON THE PHYSICAL HUMANOID. GRASP SUCCESS IS MEASURED OVER 10 TRIALS PER POSITION SPLIT, AND LANGUAGE COMPLIANCE OVER 10 TRIALS PER INSTRUCTION TYPE. GAP DENOTES THE ID-TO-OOD DROP IN GRASP SUCCESS.

	Position Generalization			Language Compliance	
Method	ID Grasp (%) ↑	OOD Grasp (%) ↑	Gap (pp) ↓	Left/Right (%) ↑	Color (%) ↑
GR00T1.7-MONO	70	40	30	100	80
GR00T-WIDE	50	40	10	0	0
GR00T	100	60	40	100	80
StereoPolicy	100	70	30	60	0
Main-XAttn	60	60	0	0	0
CVAT	100	60	40	100	80
CVAT-Flat	100	80	20	100	80
Stereo-Route	80	60	20	100	80
EATR-Stereo	**100**	**100**	0	**100**	**80**

TABLE IV
SEVERE ASYMMETRIC-OCCLUSION RECOVERY. MEAN TIME USES SUCCESSFUL TRIALS ONLY; ALL TRIALS USE A 90-S TIMEOUT.

Method	Recovery Success ↑	Mean Time (s) ↓
CVAT	30% (3/10)	> 55.0
CVAT-Flat	60% (6/10)	41.7
EATR-Stereo	**80% (8/10)**	**22.4**

gap, together with 100% grasp success on both splits, indicates that grasp performance is preserved at bottle positions absent from the demonstrations. At unseen positions, EATR-Stereo also maintains accurate bottle-directed locomotion during approach, reducing positioning errors that are a major source of subsequent grasp failures.

For semantic behavior, Table III shows 100% compliance with left/right instructions and 80% compliance with color instructions. This matches the best language-compliance performance in Table III, while providing substantially higher OOD grasp success. The stereo integration therefore improves spatial generalization without degrading language-conditioned behavior.

Occlusion recovery: We next evaluate the asymmetric-occlusion case illustrated in Fig. 3(D), where the robot hand fully occludes the bottle in the primary view but leaves it partially visible to the auxiliary camera. Each method is tested in 10 trials with a 90-s timeout. Recovery time is measured from placement at the occluded position until a successful lift and averaged over successful trials.

As reported in Table IV, EATR-Stereo recovers in 80% of trials, compared with 60% for CVAT-Flat and 30% for CVAT. Its 22.4-s mean recovery time is 46.3% lower than that of CVAT-Flat and at least 59.3% lower than that of CVAT. These results demonstrate faster and more reliable closed-loop recovery when auxiliary evidence is routed using body-structured proprioceptive state.

System-level training cost: Finally, we compare training cost and physical-task performance against the default two-image GR00T baseline and an Estimated-Depth VLA based on Depth Anything V3 [21]. All methods use the same dataset, base VLA, 60,000-step schedule, global batch size of 512, and 16 NVIDIA H20 GPUs. GR00T and EATR-Stereo keep the VLM frozen.

TABLE V
SYSTEM-LEVEL TRAINING COST AND PHYSICAL-TASK PERFORMANCE.

Method	VLM Adapt.	Time (h)	Relative	Stage (%) ↑
GR00T	Frozen	10.35	1.00×	57.5
EATR-Stereo	Frozen	**10.95**	1.06×	**80.0**
Estimated-Depth VLA	4 layers	41.80	4.04×	**80.0**

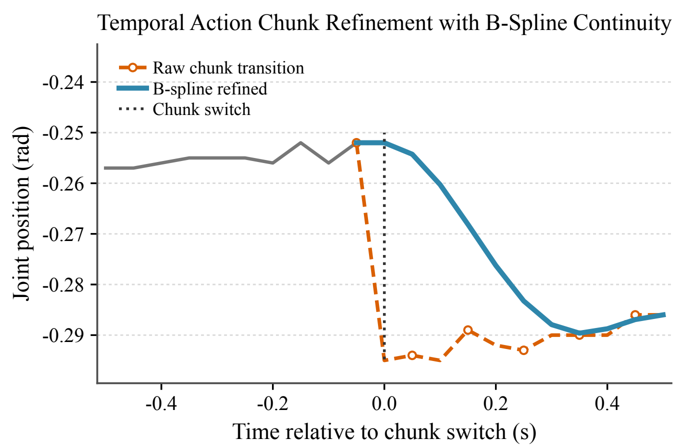

Fig. 5. Qualitative action-chunk continuity at a representative joint. The raw chunk exhibits a discontinuity at the switch, while local cubic B-spline refinement preserves boundary continuity and smoothly approaches the new trajectory.

The estimated-depth method encodes the two RGB streams separately, concatenates their features, and fine-tunes four VLM layers to align the depth features with the pretrained token space.

Table V shows that EATR-Stereo reaches an 80.0% stage success rate while requiring 10.95 h of training, only 0.60 h (5.8%) more than GR00T, whose stage success is 57.5%. The Estimated-Depth VLA also reaches 80.0% stage success but requires 41.8 h, corresponding to 4.04× the training cost of GR00T and 3.82× that of EATR-Stereo. Thus, EATR-Stereo attains the same stage-level performance as explicit estimated-depth adaptation with substantially lower training overhead.

Action-chunk continuity:

Fig. 5 shows a representative joint trajectory around an action-chunk boundary. Directly switching to the raw prediction produces an abrupt position change at the boundary, whereas local cubic B-spline refinement continues from the committed trajectory and gradually joins the new chunk over the blending window.

V. CONCLUSION

This work investigated how synchronized stereo observations can be integrated into a pretrained humanoid VLA while preserving its native primary-view representation. EATR-Stereo introduces a primary-aligned Cross-View Auxiliary Token pathway and a state-conditioned routing mechanism to selectively incorporate complementary stereo evidence without replacing the original semantic stream. Experiments on a 33-DoF humanoid with a 37-D proprioceptive state demonstrate improved long-horizon manipulation reliability, achieving 60.0% full-task success, 100.0% grasp success, and 80.0% stage success in over-100-s tasks. The proposed method also shows more reliable spatially grounded behavior during locomotion and improves recovery under severe asymmetric occlusion. Ablation studies further highlight the contributions of primary-token preservation, explicit cross-view auxiliary modeling, and structured proprioceptive context for adaptive evidence selection.

These results show that stereo observations can be effectively incorporated into existing VLA systems through lightweight representation interfaces without redesigning the pretrained vision–language backbone. EATR-Stereo enables embodiment-conditioned use of complementary visual evidence, while calibrated uncertainty modeling and explicit depth representations remain promising directions for further study. The current evaluation focuses on the Omega humanoid platform and complementary arm-based simulation benchmarks; future work will extend evaluation to larger-scale trials, diverse embodiments, and challenging stereo conditions. By combining primary-view preservation with adaptive multi-view evidence utilization, EATR-Stereo provides a practical framework for improving embodied policies under viewpoint variation and partial observability.

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